

Murtaza S. Ahmed

Undergrad Engineer | Distributed Systems, ML Systems & Infra (ML focus)

murtazaofficial@protonmail.com | +91 9390341984 | onerealti.github.io/portfolio | github.com/onerealti | www.linkedin.com/in/murahmed

Focusing on AI infrastructure, particularly GPU cluster orchestration, inference systems, model routing, resource allocation and workload scheduling.

Professional Experience

Undergraduate Research Assistant (Edge AI Systems)
Department of Mechanical Engineering (MJCET)

Dec 2025 -- Present
Hyderabad, TG

- Selected for an institution-funded R&D program under faculty supervision.
- Built a complete edge AI system on NVIDIA Jetson Nano 4GB Developer Kit, integrating an IMX719 camera and soil sensors for real-time field operation.
- Developed the on-device inference pipeline and executed real-time inference under compute and memory constraints.
- Achieved 40–70 ms latency (14–20 FPS) using TensorRT (FP16) optimization.
- Processed camera and sensor data continuously at 5–10 Hz with low-latency performance.
- Designed a stable execution loop enabling reliable operation over 4–6 hour field deployments.

Undergraduate Researcher (R&D Program)
Department of Mechanical Engineering (MJCET)

March 2026 -- Present
Hyderabad, TG

- Design and implementation of an Adaptive Multi-Objective Policy Engine for Resilient Orchestration in Heterogeneous LLM Clusters.
- Researching adaptive workload scheduling and hardware-aware resource allocation strategies for heterogeneous LLM inference systems.
- Exploring fault-tolerant orchestration approaches for distributed AI workloads across mixed compute environments.

Technical Contributor (AI & ML)
Horizon Research Group

April 2025 -- July 2025
Remote

- Contributed technical research regarding AI/ML workloads and inference performance metrics.
- Investigated distributed scheduling strategies and model routing logic for scale-out neural architectures.

Builder & Systems Architect
Verity (Decentralized Verification Platform)

July 2025 -- Present
Hyderabad, TG

- Designing and implementing a system to verify digital media provenance using blockchain and decentralized storage.
- Integrating AI modules for manipulation detection and automated claim extraction.
- Developing media provenance infrastructure using distributed storage systems including IPFS and Arweave.
- Building concurrent ingestion pipelines and immutable audit logging mechanisms for ML-based verification workflows.

Projects

Smart Agri Four-Legged Bot (SAFL-B) *(C, Python, UART/Serial, I2C, GPIO, Hardware Control)*

Dec 2025

- Awarded 1st Place at the MAKEFORHYDERABAD 2-Day Makeathon for autonomous robotics systems integration.
- Implemented deterministic UART/Serial communication protocols for multi-joint motor control.
- Developed telemetry stabilization routines to reduce terrain-induced sensor and image noise.

Technical Skills

Languages:	C, Python, Bash, Git
Systems:	Linux, Docker, Kubernetes
ML/Edge Computing:	TensorRT, NVIDIA Jetson Nano
Hardware & Interfaces:	UART/Serial, I2C, GPIO

Education

Bachelor of Engineering in Mechanical Engineering
Muffakham Jah College of Engineering & Technology (Osmania University)

May 2028
Hyderabad, TG

- Relevant Coursework: Scientific Computing, Embedded Systems Programming, System Modeling & Simulation